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[5152]-166

S.E. (Comp./IT) (II Semester) EXAMINATION, 2017

ENGINEERING MATHEMATICS—III

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Attempt Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Use of logarithmic tables, slide rule, Mollier charts, electronic pocket calculator and steam tables is allowed.

(v) Assume suitable data, if necessary.

1. (a) Solve any two : [8]

(i) $(D^2 + 1)y = x \cos x$

(ii) $(D^2 - 4D + 4)y = 8x^2 e^{2x} \sin x$

(iii) $(D^2 + 9)y = \frac{1}{1 + \sin 3x}$.

(b) Find the Fourier sine transform of $f(x) = \frac{e^{-ax}}{x}$. [4]

P.T.O.

Or

2. (a) An e.m.f. $E \sin pt$ is applied at $t = 0$ to a circuit containing a condenser C and inductance L in series the current I satisfies the equation : [4]

$$L \frac{dI}{dt} + \frac{1}{C} \int I dt = E \sin pt, \text{ where}$$

$$i = -\frac{dq}{dt}, \text{ If } p^2 = \frac{1}{LC}$$

and initially the current and the charge are zero, find current at any time t .

- (b) Find the inverse z -transform (any one) : [4]

$$(i) F(z) = \frac{z}{(z-1)(z-2)}, |z| > 2$$

$$(ii) F(z) = \frac{1}{(z-2)(z-3)}, 2 < |z| < 3$$

- (c) Solve the following difference equation to find $f(k)$: [4]

$$6f(k+2) - 5f(k+1) + f(k) = 0$$

$$f(0) = 0, f(1) = 3, k \geq 0.$$

3. (a) The first four moments of a distribution about 25 are -1.1 , 89 , -110 and 23300 . Calculate the first four moments about the mean. [4]

- (b) In a Poisson distribution if : [4]

$$P(r = 1) = 2 P(r = 2)$$

then show that :

$$P(r = 3) = 0.0613.$$

- (c) Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at $(2, -1, 2)$ towards the point $\bar{i} + \bar{j} - \bar{k}$. [4]

Or

4. (a) Attempt any one : [4]

(i) For scalar functions ϕ and ψ , show that :

$$\nabla \cdot (\phi \nabla \psi - \psi \nabla \phi) = \phi \nabla^2 \psi - \psi \nabla^2 \phi.$$

(ii) Show that :

$$\nabla^2 \left(\frac{\bar{a} \cdot \bar{b}}{r} \right) = 0.$$

- (b) Show that the vector field :

$$\bar{F} = (ye^{xy} \cos z) \bar{i} + (xe^{xy} \cos z) \bar{j} + (-e^{-xy} \sin z) \bar{k}$$

is irrotational. Also find the corresponding scalar ϕ , such that $\bar{F} = \nabla \phi$. [4]

- (c) If the two lines of regression are $9x + y - \lambda = 0$ and $4x + y - \mu = 0$ and the means of x and y are 2 and -3 respectively, then find λ , μ and the coefficient of correlation between x and y . [4]

5. (a) Apply Green's Lemma to evaluate the : [5]

$$\oint (3x^2 - 8y^2) dx + (4y - 6xy) dy,$$

where C is the boundary of the region defined by $y = \sqrt{x}$, $y = x^2$ in the plane $z = 0$.

(b) If : [4]

$$\vec{F} = (x^2 + y - 4)\hat{i} + 3xy\hat{j} + (2xz + z^2)\hat{k},$$

evaluate :

$$\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, dS,$$

where S is the surface of the sphere :

$$x^2 + y^2 + z^2 = 16$$

above the xy plane.

(c) Evaluate : [4]

$$\iint_S \vec{F} \cdot \hat{n} \, dS,$$

where

$$\vec{F} = (2x + 3y^2z^2)\hat{i} - (x^2z^2 + y)\hat{j} + (y^3 + 2z)\hat{k}$$

and S is the surface of the sphere with centre (3, -1, 2) and radius 3.

Or

6. (a) Evaluate $\int_C \vec{F} \cdot d\vec{r}$ from the point (0, 0, 0) to (1, 1, 1) along the curve $x = t, y = t^2, z = t^3$, given : [4]

$$\vec{F} = xy\hat{i} - z^2\hat{j} + xyz\hat{k}.$$

(b) Using divergence theorem, evaluate $\int_S \vec{F} \cdot \hat{n} \, dS$ over S, the surface of unit cube bounded by the co-ordinates planes and the planes $x = 1, y = 1$ and $z = 1$ where $\vec{F} = 2x\hat{i} + 3y\hat{j} + 4z\hat{k}$. [4]

- (c) Apply Stokes' theorem to evaluate $\int_C \bar{F} \cdot d\bar{r}$, where $\bar{F} = y\hat{i} + z\hat{j} + x\hat{k}$, where C is the curve given by :

[5]

$$x^2 + y^2 + z^2 - 2ax - 2ay = 0$$

and

$$x + y = 2a.$$

7. (a) If $v = 4xy(x^2 - y^2)$, find u such that $f(z) = u + iv$ is analytic and determine $f(z)$ in terms of z . [4]
- (b) Evaluate $\oint_C \tan z \, dz$ where C is the circle $|z| = 2$. [5]
- (c) Show that the transformation $W = \frac{1}{z}$ maps the circle $x^2 + y^2 - 6x = 0$ onto a straight line in W-plane. [4]

Or

8. (a) If $f(z) = u + iv$ is analytic and $u + v = \sin x \cdot \cosh y + \cos x \cdot \sinh y$, then find $f(z)$ in terms of z . [4]
- (b) Evaluate $\oint \frac{e^z}{(z-1)^2(z-2)} dz$ where 'C' is the contour $|z-2| = \frac{3}{2}$ by using Cauchy's residue theorem. [5]
- (c) Find the bilinear transformation which maps the points $0, \frac{1}{2}, 1+i$ from z -plane into the points $-4, \infty, 2-2i$. [4]